Homework 6: Thin Films

Constants used. $k = 1.381 \times 10^{-23}$ J/K $= 8.617 \times 10^{-5}$ eV/K; 1 amu $= 1.661 \times 10^{-27}$ kg; $N_A = 6.022 \times 10^{23}$ mol^{-1}; 1 torr = 133.32 Pa.

Vapor pressures. These were read from the vapor-pressure chart attached to the homework (Alcock et al.):

Vapor pressure (torr)	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}	3×10^{-2}
Al	829 °C	901 °C	999 °C	1115 °C	1248 °C	1321 °C
Au	958 °C	1041 °C	1151 °C	1281 °C	1429 °C	1511 °C

Between these points, log P is interpolated linearly in 1/T (Clausius–Clapeyron).

1. Crucible temperature for equal Al and water-vapor fluxes (10 pts)

Check against your textbook: I couldn't access the text of Campbell's Example 12.2. I assumed its usual setup: a background pressure of 10^{-6} torr of water vapor at room temperature (300 K), and the evaporator from Problem 2 (crucible area A = 10 cm^2, wafers on a sphere of radius $r_0 = 50$ cm). If the example uses different values, only the temperature changes; see the table at the end of this question. The growth rate does not depend on the geometry.

Flux of gas molecules striking a surface (Campbell, Eq. 12.2):

$$\Phi = P / \sqrt{(2\pi mkT)} = 3.51 \times 10^{22} \cdot P / \sqrt{(M \cdot T)} \quad [\text{molecules}/(\text{cm}^2 \cdot \text{s}),\ P \text{ in torr},\ M \text{ in amu},\ T \text{ in K}]$$

Water-vapor flux on the wafer (M = 18, T = 300 K, $P = 10^{-6}$ torr):

$$\Phi_{H_2O} = 3.51 \times 10^{22} \times 10^{-6} / \sqrt{(18 \times 300)} = \mathbf{4.78 \times 10^{14}\ molecules/(cm^2 \cdot s)}$$

Al flux arriving at the wafer. For wafers on the deposition sphere (see Q4), the flux leaving the source, $\Phi_s \cdot A$, spreads evenly over the sphere's area $4\pi r_0^2$:

$$\Phi_{\text{Al, wafer}} = \Phi_s \cdot A / (4\pi r_0^2), \quad \text{where } \Phi_s = 3.51 \times 10^{22} \cdot P_e / \sqrt{(27 \cdot T)}$$

Set the two fluxes equal. The flux needed at the source is:

$$\Phi_s = \Phi_{H_2O} \cdot 4\pi r_0^2 / A = 4.78 \times 10^{14} \times 4\pi (50\ \text{cm})^2 / (10\ \text{cm}^2) = 1.50 \times 10^{18}\ \text{atoms}/(\text{cm}^2 \cdot \text{s})$$

$$P_e = \Phi_s \cdot \sqrt{(27\ T)} / 3.51 \times 10^{22}$$

This has to be solved together with the Al vapor-pressure curve, because P_e depends on T. Iterating: guessing T ≈ 1500 K gives $P_e \approx 8.6 \times 10^{-3}$ torr. The chart says Al reaches 8.6×10^{-3} torr at about 1239 °C (1512 K), which is consistent with the guess.

Answer: the crucible must be at **T ≈ 1240 °C (≈ 1510 K)**, where P_e(Al) ≈ 9×10^{-3} torr.

Growth rate. At equal fluxes, Al atoms arrive at $\Phi = 4.78 \times 10^{14}$ atoms/(cm^2·s) and stick. The atomic density of Al is:

$$n_{Al} = \rho N_A / M = (2.70\ \text{g/cm}^3)(6.022 \times 10^{23})/(26.98\ \text{g/mol}) = 6.03 \times 10^{22}\ \text{atoms/cm}^3$$

$$\text{Growth rate} = \Phi / n_{Al} = 4.78 \times 10^{14} / 6.03 \times 10^{22} = 7.9 \times 10^{-9}\ \text{cm/s}$$

> **Answer:** growth rate ≈ **0.8 Å/s ≈ 48 Å/min (about 4.8 nm/min)**.

This is slow, and at this rate there is about one water molecule striking the wafer for every Al atom. That is the point of the example: at 10^{-6} torr, a slow Al deposition can take up a lot of oxygen and hydrogen. A faster rate (hotter crucible) or a lower base pressure is needed for clean films.

Example 12.2 geometry (if different)	Required source flux	P_e(Al)	Crucible T
A = 10 cm^2, r_0 = 50 cm (used above)	1.5×10^{18} /(cm^2·s)	8.6×10^{-3} torr	≈ 1240 °C
A = 10 cm^2, r_0 = 30 cm	5.4×10^{17} /(cm^2·s)	3.0×10^{-3} torr	≈ 1175 °C

In every case the growth rate is still ≈ 0.8 Å/s, because it is fixed by the water flux.

2. Au/Al co-evaporation with a rotating shutter (10 pts)

Timing. At 10 rpm one revolution takes 6 s. The opening is half the shutter, so each source deposits for **3 s per cycle**. One 3 Å monolayer per cycle therefore needs a deposition rate, while the source is uncovered, of:

$$R_d = 3\ \text{Å} / 3\ \text{s} = 1\ \text{Å/s} = 1 \times 10^{-10}\ \text{m/s} \quad \text{(for both Au and Al)}$$

Required evaporation rate. On a deposition sphere of radius r_0 the deposition is uniform (Q4): $R_d = R_{ev} / (4\pi r_0^2 \rho)$. So

$$R_{ev} = 4\pi r_0^2\, \rho\, R_d, \quad \text{with } 4\pi r_0^2 = 4\pi(0.50\ \text{m})^2 = 3.14\ \text{m}^2$$

- Au: $R_{ev} = 3.14 \times 18{,}900 \times 10^{-10}$ = **5.94×10^{-6} kg/s**
- Al: $R_{ev} = 3.14 \times 2{,}700 \times 10^{-10}$ = **8.48×10^{-7} kg/s**

Required vapor pressure (Hertz–Knudsen, A = 10 cm^2 = 10^{-3} m^2):

$$R_{ev} = A \cdot \sqrt{(m / 2\pi kT)} \cdot P_e \implies P_e = R_{ev} / [A \sqrt{(m / 2\pi kT)}]$$

Solve this together with each vapor-pressure curve, as in Q1:

	Atom mass m	Required P_e	Source temperature (from the chart)
Gold	197 amu = 3.27×10^{-25} kg	4.1 Pa = 3.1×10^{-2} torr	**≈ 1510 °C (≈ 1780 K)**
Aluminum	27 amu = 4.48×10^{-26} kg	1.46 Pa = 1.1×10^{-2} torr	**≈ 1250 °C (≈ 1530 K)**

Check for Au: $\sqrt{(m/2\pi kT)} = \sqrt{(3.27 \times 10^{-25} / (2\pi \times 1.381 \times 10^{-23} \times 1782))} = 1.46 \times 10^{-3}$ s/m, so $P_e = 5.94 \times 10^{-6} / (10^{-3} \times 1.46 \times 10^{-3}) = 4.1$ Pa. ✔

Answer: run the **Au source at about 1510 °C** and the **Al source at about 1250 °C**. Each then deposits 1 Å/s while uncovered, which is one 3 Å layer per shutter revolution.

Gold needs a much hotter source for two reasons. Matching the same thickness rate takes 7 times more mass per second (its density is 7 times higher). And gold's vapor pressure is lower than aluminum's at any given temperature. Equal thicknesses also give a nearly 50/50 atomic mixture: Au has 5.78×10^{22} atoms/cm^3 and Al has 6.03×10^{22}, a 49:51 ratio.

3. Reaction-rate-limited deposition: 700 °C → 800 °C (10 pts)

In the reaction-rate-limited regime, the deposition rate follows an Arrhenius law, $R \propto \exp(-E_a/kT)$. So

$$R(800\ °C) / R(700\ °C) = \exp[\ (E_a/k)(1/T_1 - 1/T_2)\ ]$$

$$= \exp[\ (2\ eV / 8.617 \times 10^{-5}\ eV/K)(1/973\ K - 1/1073\ K)\ ] = \exp(23{,}210 \times 9.58 \times 10^{-5}) = \exp(2.22) \approx 9.2$$

Prediction: R(800 °C) ≈ 9.2 × 1000 Å/min ≈ **9,200 Å/min (about 0.9 µm/min)**.

If the measured rate is much lower: the process has most likely moved into the **mass-transport-limited** (gas-phase diffusion-limited) regime. The surface reaction has become so fast that the rate is now set by how quickly reactant gas can diffuse through the boundary layer to the wafer. Diffusion depends only weakly on temperature (roughly $T^{1.5–2}$), so the rate no longer increases exponentially. Other possibilities are depletion of the reactant along the reactor, or gas-phase reactions (homogeneous nucleation) using up reactant before it reaches the wafer.

How to prove it:

1. **Arrhenius plot.** Measure the rate at several temperatures between 600 and 900 °C and plot ln(rate) against 1/T. A reaction-limited process gives a straight line with slope $-E_a/k$ (−2 eV). If the slope flattens above some temperature, the process has become transport-limited there.
2. **Change the gas flow and pressure at 800 °C.** A transport-limited rate increases with gas velocity, since a thinner boundary layer ($\delta \propto 1/\sqrt{v}$) lets reactant reach the wafer faster. It also changes with total pressure and with the reactant's partial pressure. A reaction-limited rate does not depend on flow.
3. **Check uniformity.** In the transport-limited regime, the rate depends on wafer position and spacing. Wafers further downstream, or more tightly packed, grow thinner films. In the reaction-limited regime, all wafers at the same temperature grow at the same rate.

4. Uniform deposition on a sphere (10 pts)

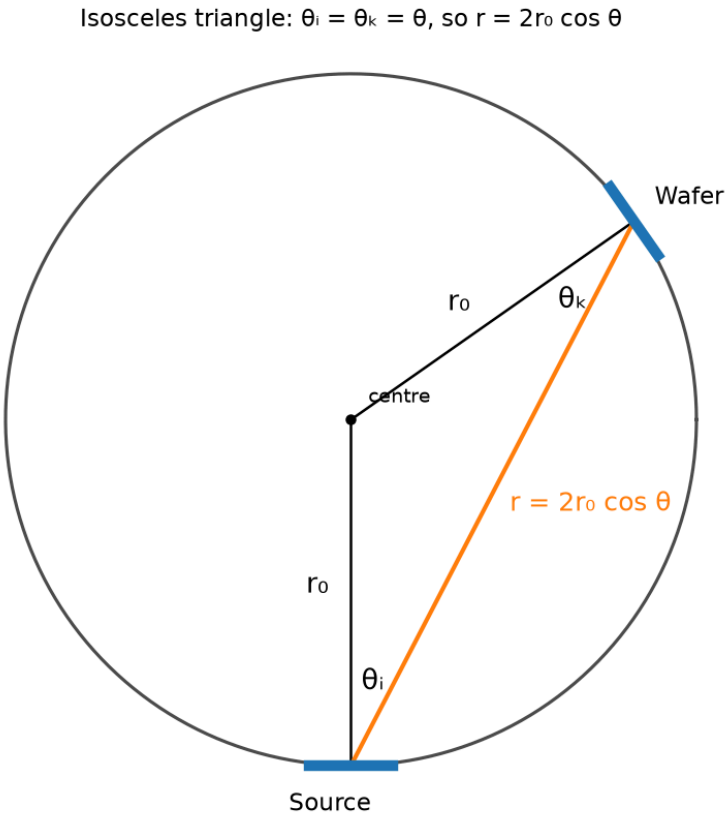


Figure 1. A small planar source and a wafer, both on the inside of a sphere of radius r_0 and both facing its centre.

Emission from a small planar source (Knudsen cosine law). The source loses mass at a total rate R_{ev}. The mass sent into a small solid angle dω at angle θ_i from the source's normal is:

$$dR = (R_{ev} / \pi) \cos \theta_i \, d\omega$$

(The 1/π normalizes it: ∫ cos θ dω over the half-space in front of the source equals π.)

Mass arriving at the wafer. A wafer area element dA_k at distance r, whose normal makes angle θ_k with the line to the source, covers a solid angle $d\omega = \cos \theta_k \, dA_k / r^2$. So the mass deposition rate per unit wafer area is:

$$dR / dA_k = (R_{ev} / \pi) \cdot \cos \theta_i \cos \theta_k / r^2$$

Geometry on the sphere. The source and the wafer both lie on the sphere and face its centre, so each normal points along a radius. The triangle formed by the source, the wafer and the centre has two sides equal to r_0, so it is isosceles. Its base angles are equal, so $\boldsymbol{\theta_i = \theta_k = \theta}$. Dropping a perpendicular from the centre to the chord splits the chord in half: $r/2 = r_0 \cos \theta$, so

$$r = 2r_0 \cos \theta$$

Substitute:

$$dR / dA_k = (R_{ev} / \pi) \cdot \cos^2\theta / (4r_0^2 \cos^2\theta) = R_{ev} / (4\pi r_0^2)$$

The $\cos^2\theta$ terms cancel, so the result **does not depend on θ**, the wafer's position on the sphere. The thickness growth rate is

$$R_d = R_{ev} / (4\pi r_0^2 \rho)$$

This is the same for every wafer on the sphere. ∎

Physically, wafers far from the source are farther away (r is larger), but they face the source more squarely and the source sends more material toward them, and these effects exactly balance. This is why planetary evaporators mount wafers on a spherical dome that passes through the source. The result assumes a small planar source with an ideal cosine emission pattern. Real e-beam and boat sources are somewhat more directional, so planetary rotation is also used.

Sources

- S. A. Campbell, *Fabrication Engineering at the Micro- and Nanoscale*, Ch. 12 (Physical Deposition: Evaporation and Sputtering), Eq. 12.2 and the deposition-rate equations.
- Vapor-pressure chart attached to the homework (data adapted from Alcock et al.). The values were read off the Al and Au curves.
- Kurt J. Lesker Company, aluminum evaporation notes (cross-check: Al at 10^{-4} torr ≈ 1010 °C): https://www.lesker.com/newweb/deposition_materials/deposition-materials-notes.cfm?pgid=al1